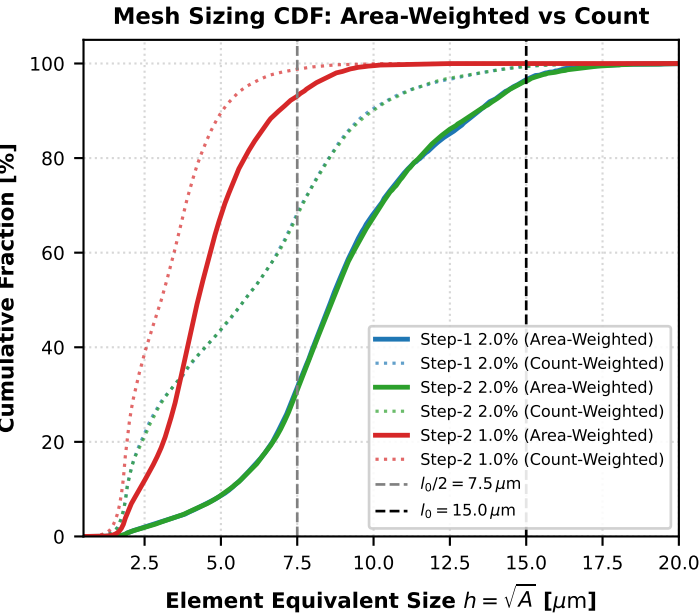
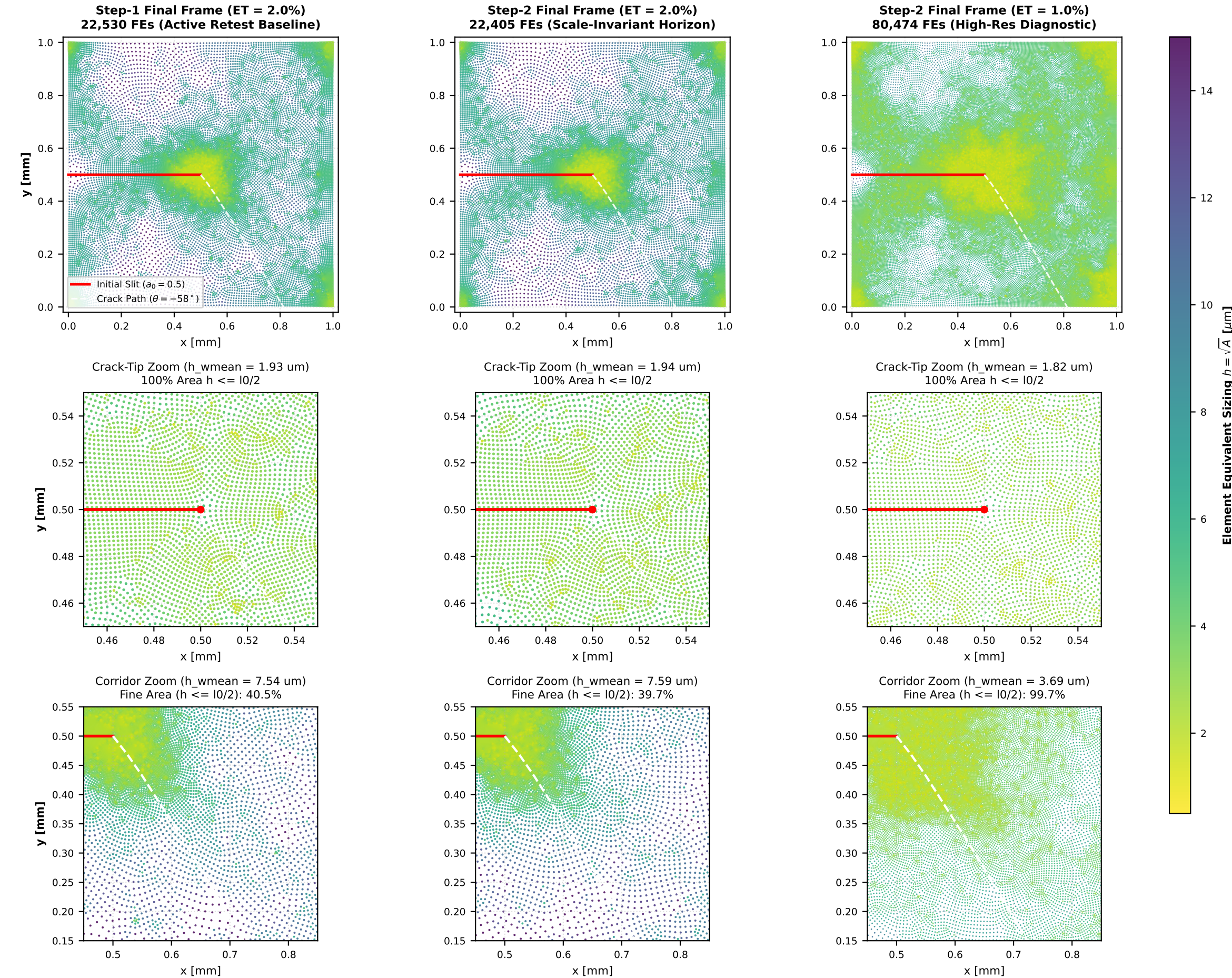


Mode-II Gate M2-3: Three-Way Adaptive Discretization & Area-Weighted Refinement Verification
Pandey & Kumar (2025) Pure Shear Specimen ($l_0 = 15.0\mu\text{m}$, $\Omega = 1.0 \times 1.0\text{ mm}$)



Metric / Discretization	Step-1 (ET=2.0%)	Step-2 (ET=2.0%)	Step-2 (ET=1.0%)	Scientific Interpretation
Finite Element Count	22,530	22,405 (-0.55%)	80,474 (+259%)	Scale invariance (2%) vs extreme expansion (1%)
Minimum Size h_{min} [μm]	0.73 (0.049 l_0)	0.76 (0.051 l_0)	0.60 (0.040 l_0)	All meshes resolve crack singularity ($h_{min} \ll l_0$)
Mean Size (Count) [μm]	5.83 (0.389 l_0)	5.85 (0.390 l_0)	3.23 (0.216 l_0)	Count mean biased by high density of small elements
Mean Size (Area-Wtd) [μm]	8.96 (0.598 l_0)	8.99 (0.599 l_0)	4.50 (0.300 l_0)	Reflects true spatial domain grading across specimen
Domain Area $h \leq l_0/2$ [%]	31.54%	30.96%	93.09%	1% mesh refines 93.1% of entire domain (near-global)
Tip Area $h \leq l_0/2$ [%]	100.0%	100.0%	100.0%	100% fine coverage around notch tip across all 3 meshes
Corridor Area $h \leq l_0/2$ [%]	40.5%	39.7%	99.7%	1% covers 99.7% of shear box; 2% covers 40% selectively
Far-Field Area $h \geq l_0$ [%]	4.60%	4.76%	0.00%	1% completely eliminates far-field coarsening (0.0% $\geq l_0$)
Scientific Role & Status	Active Retest Baseline (Job 1411103)	Equivalent Production Candidate (NOT YET PROVEN)	High-Cost Diagnostic (Unqualified for Production)	2% meshes optimal for thesis; 1% diagnostic only